Bandits for Online Exploration in Deep Recommenders

CS310 Third Year Project: Progress Report Sulay Chaudhry (5567816)

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1 Introduction

Recommender systems (RS) are crucial in the digital era not only from the historical perspective of e-commerce and advertising, but as tools that fundamentally shape the personalised user experiences of billions of users in the era of information overload [1]. With the rise of social media – and as the line between recommendation, information discovery, and information retrieval (IR) has continued to blur – recommenders now play an *interventional* [2] role in shaping both public access to and attitudes towards digital information, rather than a purely *observational* one.

Traditional evaluation metrics for recommender models, as predominantly batch-trained supervised learning models [3], have been centred around their ability to maximise for reward with respect to historical user-item interaction patterns. This has been demonstrated to lead to *Matthew effect*-style [4] degenerative suboptimal feedback loops in which recommender models focus on policies that reinforce existing candidates with high existing engagement [5]. Additional to global biases, there is the opportunity for local biases in which hyperpersonalised algorithms may have adequate coverage and confidence of the recommendation space, but fail to cater for user-specific content diversity [6]; this is particularly relevant in discussions around the noted *filter bubble* effect [7], in which RS algorithms can reinforce individual biases through limited content exposure.

These biases continue to be seen in the latest industry-standard generation of deep-learning recommenders [3, 6, 8], chosen predominantly as they excel in discovering complex non-linear relationships between users and items at incredibly large scale. This *exploitation*-centred model, in which models are batch-trained to align the recommendation policy with a large historical set of known user-item interactions, often fails to capture the importance of models to maximise for metrics other than immediate prediction accuracy [9, 10].

Recent work within recommenders has focused on non-accuracy metrics [11, 6], especially within much more complex multi-stage recommenders, to begin to alleviate such concerns. However they often still are batch-trained offline without a system built for purpose to guide and adapt exploration in real-time on a per-user basis: the exploration policy is often not an online adaptive policy, but one that is part of a process that continues to have a delayed feedback loop [3].

Contextual bandits are a form of reinforcement learning that have gradually picked up significant attention in the RS space [12, 13, 1], with applications across personalised re-ranking and noted application in high-adaptability recommendation contexts requiring rigorous exploration

such as news recommendation [14]. They are particularly relevant as they model the *exploration-exploitation tradeoff* in real-time, optimising relative to both reward and uncertainty to continually optimise exploration of the user's preferences. Their use in this vein predates the modern (post-2016) rise of modern RS based on multi-stage deep ensemble models [15, 16]; however they historically faced major issues with scalability [17], offline evaluation [18], and the limited representation power of their (predominantly) linear models [19] in comparison.

In recent years, however, a renewed interest in bandits as viable candidates for directly dealing with the tradeoff has been studied [17, 20]. Advances in off-policy evaluation have made it significantly easier to make meaningful offline claims for RL (especially bandit) systems [18, 2] – a major issue historically that prevented their application, restricting system design to online testing or bias-prone forms of offline evaluation. Additionally, research has outlined the novel viability of an effective scalable *hybrid* approach [3, 20] leveraging a mix of deep supervised (batch-trained offline) RS as a base embedding model, with bandits (adapt online in real-time) as part of a supplementary layer ensuring real-time policy adaptation balancing exploration and exploitation.

This research project centres around the development of such an offline-online hybrid model. In particular, it directly aims to demonstrate the benefits of supplementing an industry-standard, well-documented deep supervised RS architecture – the two-tower model [21, 22, 16] – with three dominant contextual bandit algorithms across not only immediate recommendation quality as it adapts to user feedback in real-time, but additionally across beyond-accuracy metrics such as the Gini index and long-tail exposure as measures of popularity bias [11]. Such work aims to supplement and reinforce the renewed interest in RL's real-time exploration as a viable supplementary option for improving the quality of existing RS [1].

2 Background Research

2.1 Supervised Learning in Recommender Systems

Historically, the dominant RS architectures have been centred around *supervised learning* due to the abundance of large logged user-item interaction histories for use in learning complex embedding and ranking relationships [3, 17], formulating recommender systems as an *observational* [2] problem situated in an environment in which the recommender must continually know more about the users and items. This has proved to be a highly effective approach to recommendation, and has continually lent itself well to large-scale offline evaluation. The two central supervised techniques have been *collaborative filtering* and *content-based filtering* [23, 16];

both highly successful in their own right, and now usually used in tandem alongside other approaches.

Collaborative filtering is built around the central heuristic that *similar users like similar items*: historical data is used to learn complex relationships between massive sets of users and items, such that purely through interaction data an embedding space is learned by the models [24]. It reduces the recommendation problem to one of simple vector similarity or nearest neighbours search, given that the embedding space is accurate.

Content-based filtering is built around an information retrieval approach: rather than centering its understanding of users and items wholly on interaction data, it uses underlying methods to extract metadata (features) for the candidate items, and then aims to learn what kinds of item features this user specifically looks for [23]. This builds up a user profile of their preferences, given a robust foreknowledge of the item's qualities.

The mid-2010s saw the rise of deep-learning recommender systems as the dominant paradigm [16]: deep-learning approaches to collaborative filtering enabled the learning of complex non-linear interaction patterns, while content-based filtering gradually grew into learning complex meaningful embeddings for items through deep learning models on text, image, and video. Beyond this, the rich ability of deep-learning models to collate all of this data while learning complex non-linear relationships – while inherently learning the importance of different factors – has led to the state-of-the-art: hybrid models such as *two-tower* [21, 25, 22] which learn complex user and item embeddings that incorporate both feature-engineered inputs and user-item interaction history.

Two-tower (dual encoder) is specifically built around two separate neural networks: one that learns a *user embedding* and one that learns an *item embedding* such that they operate within a shared embedding space, such that compatible pairs (of items, users, or user-items) have high similarity, and incompatible pairs generally have low similarity [21]. It has become the industry-standard for candidate generation, representation learning, and retrieval [16]; much of recent work is centred around reducing its biases [25, 22].

Graph Neural Networks (GNNs) have picked up more traction in recent years as a competitive alternative for scalable embedding learning and candidate generation if utilised correctly [26], and this project's model will aim to be generalisable such that a GNN can replace the two-tower in future work, but two-tower still continues to have widespread use as a highly versatile state-of-the-art architecture [16], and GNNs often are used as a supplement to two-tower rather than a direct replacement. As noted in the specification (Appendix A), the aim of this work is not

to beat the state-of-the-art deep-learning models, but instead to demonstrate the viability of a modern hybrid bandit architecture, for which two-tower contains known literature to provide the groundwork [3].

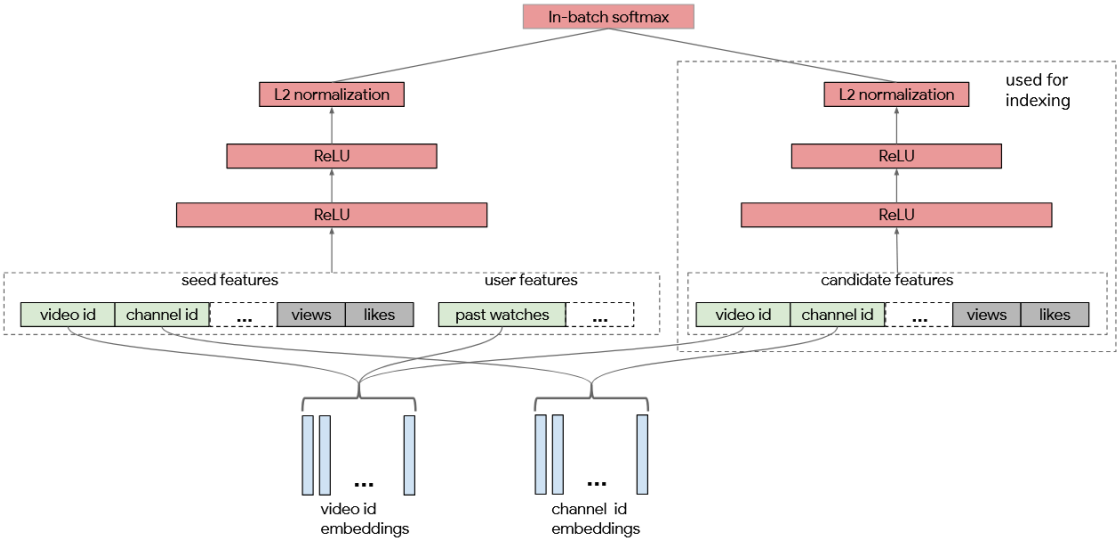


Figure 1: A two-tower model designed for YouTube [21]

2.2 Contextual Bandits in Recommender Systems

The multi-armed bandit (MAB) is a reinforcement learning problem in which an agent (*learner*) has a fixed set of actions $\mathcal{A}$ (*arms*) to pick between, without underlying knowledge of the reward structures of these arms. The learner interacts sequentially with the environment, revealing at round t a reward $X_t \in \mathbb{R}$. The common objective is to maximise cumulative reward across the *horizon* n of total rounds played: crucial to this is the *exploration-exploitation tradeoff*, in which sufficient exploration is required to identify the best arm to pick, but excessive exploration leads to wasted rounds in which the optimal arm is not played [27].

The inherent balance of exploration and exploitation inherent to this problem has made it a widely adopted contender for exploration-aware recommendation systems [1, 13]: recommendation candidates are treated as arms, and the environment dictating the rewards of those arms is the user. To meaningfully deal with large user and item spaces, *structure* assumptions are imposed such that it is assumed that cross-user and cross-item (collaborative-style) learning is possible through their representation within a *context* feature vector; such is the *contextual* bandit.

The major contextual bandits seen in recommenders are *linear bandits* [12, 28], assuming that reward can be modelled as a linear function of these user and item features, with three dominant

algorithms: *LinUCB* [14, 29], *Thompson Sampling* (TS) [30, 31], and ε-greedy [27]. LinUCB and TS are both high-performance contextual bandit algorithms, but differ in their assumptions: LinUCB is a deterministic algorithm that models uncertainty continually for arms, while TS is a stochastic algorithm that continually samples from continually updated distribution estimates in a Bayesian manner. Their comparable application typically depends on the specific nature of the recommendation environment: LinUCB is often seen to converge to higher rewards in a stable immediate-reward environment [13], but TS often outperforms in unstable delayed-reward environments due to its stochastic uncertainty model [12].

Recent work extends beyond these linear models, using deep neural networks to learn non-linear reward mappings such as *NeuralUCB* [32]; they are beyond the work of this project due to their complexity, and due to major noted issues with computational complexity [17]. Other work, such as the *Neural-Linear* approaches [33, 19], directly inform this project through their exploration of deep neural networks as effective means of context embedding learning; this mirrors the role of deep neural networks within the proposed hybrid system.

3 Progress Overview

3.1 Project Status and Changes

All literature review, research, and system design work for the recommender is complete; the finalised system outline is seen in Section 3.2. This took considerably longer than expected by the project timetable seen in the specification (Appendix A) due to various factors.

A major unexpected factor was the significant work required to understand the recommender system landscape, especially on both the *deep-learning* supervised recommender side and the *contextual bandit* side; additionally, the novel nature of these hybrid systems posed a challenge for ensuring that the system design was well-backed by literature while still being a meaningful architecture feasible by an undergraduate. This was resolved through extending this research and system design period, as the project was at risk of overcommitting to a system design without any clear indication of feasibility.

Additionally, the original specification is overly optimistic in attempting to tackle three central concerns of *scalability*, *recommendation quality* and *exploration quality* at once; an analysis of the literature firstly shows that scalability for feasible industry-scale bandit systems now either involves complex custom neural implementations that are not feasible without much more background [17] or significant additional system complexity [20], and that proper scalability con-

cerns are a difficult task with the most influential work done in-house on real industry-level scales at companies like YouTube [3] – as such, industry-level scalability has been deprioritised, and future work can extend this project through the use of these industry-standard solutions.

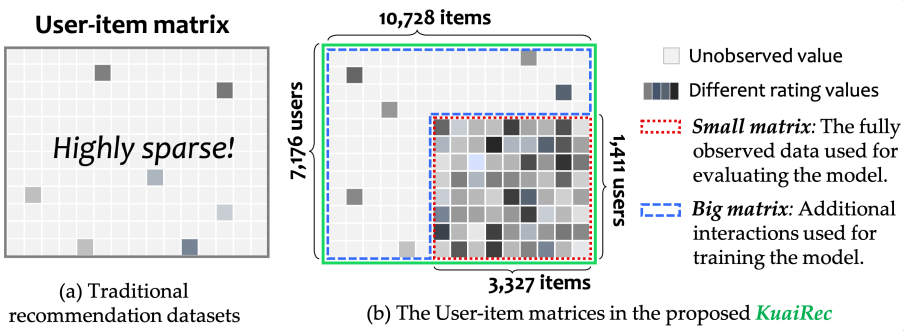
Other circumstances render the original timetable inaccurate. One is that the timetable did not account for a significantly higher workload in the first term: four modules were taken in Term 1, as opposed to two in Term 2; applications and in-person interviews additionally affected time management; two of the modules (Compiler Design and Machine Learning) had large courseworks spanning coding, report writing, and video that took weeks to complete; and mitigating circumstances that affected both the Christmas buffer period and the new term.

The choice was additionally made to move dataset from MovieLens [34] to KuaiRec [35]; this is due to the fact that further research around *off-policy evaluation* found that simulating recommendations that users were not actually presented with requires rigorous *counterfactual* [2, 18] methods to attempt to predict around expected reward; KuaiRec, on the other hand, contains an almost-fully observed dataset for offline recommender simulation. This has simplified the evaluation process massively, while massively improving the prospective credibility of this project's results in mitigating counterfactual bias considerations.

Due to the completion of the system design – including the resolving of many major bridging ambiguities between the components – and significant time spent to gain familiarity with `PyTorch`, the *two-tower architecture* and *contextual linear bandits*, the project remains feasible especially in its clarified form restricting scope and preventing additional scope creep. Additionally, iterative redesign of the system will not be required in the same way, as the final design has been rigorously aligned with and justified through standards seen in literature.

3.2 System Outline

KuaiRec [35] has been chosen as the dataset for the evaluation of this recommender; this is due to aforementioned considerations around off-policy evaluation. As outlined in section 2, KuaiRec's almost fully-observed interaction matrix (99.7% density) enables thorough evaluation of an unrestricted policy for the hybrid recommender without relying on counterfactual evaluation. KuaiRec's *big matrix* will be used to train the two-tower embedding model using large (but sparse) historical user-item data, as standard and realistic for a deployed recommender; the *small matrix* will be used to directly simulate the bandit policies, with explicit separation to ensure that data from the latter does not leak into the former.



(a) Traditional recommendation datasets

(b) The User-item matrices in the proposed *KuaiRec*

Figure 2: KuaiRec's user-item interaction matrices [36]

Work will be centred around the immediately available `big_matrix.csv` and `small_matrix.csv` features – most importantly `user_id`, `video_id`, and `watch_ratio`; these will be supplemented with KuaiRec's large amount of side information to inform embedding learning where necessary – most importantly the item category tags `item_categories.csv` and the anonymised user features in `user_features.csv`.

These features will be fed into the chosen deep embedding learner – a **two-tower** [21, 22] (dual encoder) model, chosen due to its high expressive power in learning complex representations of users and items given its relative simplicity in implementation and its continued status as the industry-standard state-of-the-art baseline for embedding and retrieval [26, 16]. User-specific features will form the input layer of the *user tower* and item-specific features will form the input layer of the *item tower*. At inference time, the similarity between a user embedding and an item embedding is as simple as calculating their dot product.

The baseline tower architectures will be that of two hidden fully-connected RELU-activated layers of sizes 1024 and 512, before a final fully-connected output layer of size 128, matching a standard baseline used at YouTube [21] that mirrors closely that of other high-quality tower implementations [15, 22]. Training will match the standard in-batch negative sampling training process seen in literature for two-tower [15, 21]: this efficiently leverages items that a user did not interact with as examples of user-item interactions to minimise similarity for, ensuring that users and items are sufficiently spaced within the global embedding space in favour of their positive user-item interactions. This will be done through *cross-entropy loss* on the user-item dot products as is easily available and standard in PyTorch [37].

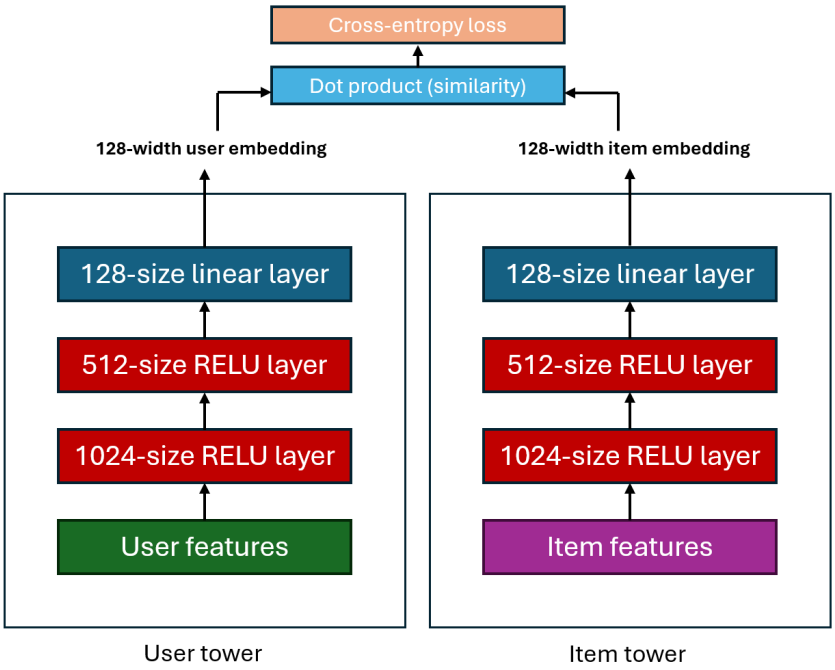


Figure 3: Architecture of the two-tower model

Lattimore's framing of a *feature map* $\psi(c, a) \in \mathbb{R}^d$ [27, p. 239] for mapping context (user embedding vector) c and action (item embedding vector) a representations to a combined feature vector will guide the understanding of using the outputs of this two-tower model for a **contextual linear bandit** [27, p. 240]. The bandit aims to learn a linear function on the d-dimensional feature vector through a weight θ_* such that reward is modelled as $\langle \theta_*, \psi(c, a) \rangle$ alongside additional stochastic noise η.

The implementation of ψ could be such that it is the simple concatenation of the user embedding c (context) from the user tower and item embedding a (action) from the item tower; concatenating additionally the element-wise multiplication $c \odot a$ has been noted however as a simple, but effective, means of allowing the linear model to learn cross relationships between the user and item embeddings [17]. As such, the contextual bandit will receive as its feature vector input the vector $\mathrm{concat}(c, a, c \odot a)$; a full outer product is avoided due to its large computational complexity, and to avoid overfitting of the linear model.

The baseline for evaluation will be that of a *purely greedy* output from the two-tower model – that is the recommendation with highest similarity purely based on the embeddings, or for a fixed user embedding u the item with embedding i such that $u \cdot i$ is maximised; this is equivalent to *maximum inner-product search* [38] for which the relatively small size of KuaiRec in comparison to industry-scale item catalogues means that even a brute-force exact solution is computationally

feasible for evaluation without the need for efficient (potentially non-exact) algorithms.

The three central algorithms for the contextual linear bandit will be **LinUCB** [14, 29], **Thompson Sampling (TS)** [30, 31], and **ε-greedy** [27, p. 95] as the "traditional" bandit algorithms used widely within industry, especially in conjunction with supervised methods [13, 12]. As discussed prior, the role of ε-greedy is that of injecting fixed non-user-specific exploration into a system; it is a simple baseline that can demonstrate even the benefits of even the smallest amount of exploration within a hybrid recommender system. LinUCB and TS are expected to perform significantly better on both exploration metrics and in analysing user reward due to their adaptive means of exploration discussed further in Section 2.

After the user-item embedding towers are trained on the big matrix, model simulation will operate using the small matrix; the recommendation candidates become the set of 3327 items, and the set of users to evaluate for becomes the 1411 users, both for which this matrix can non-counterfactually be evaluated offline for any policy due to the 99.7% interaction log density [35]. A random stream of users invoking the recommender will be simulated: for each time step t, a random user ID will be sampled and the system will simulate all four (the greedy base and the three contextual bandit) algorithms to determine a chosen item to serve the user for each model. The ground truth user-item interaction is then observed and the models are updated before progressing to the next time step.

An important consideration noted by the KuaiRec dataset is that it does not contain a binary "the user liked this item" signal; they note that this is due to like sparsity [35]. As such, the positive reward signal must be chosen and imposed upon the dataset; this will be a notable parameter to consider during implementation, but the paper notes an example binary baseline `watch_ratio >= 2.0` (indicating that a user has watched the video twice) as a strong indicator that the user liked the video. This will be used initially and tweaked as required; an interesting consideration is whether it is feasible (or informative for the recommender) to model a robust continuous notion of reward from this watch ratio beyond a binary signal, although the implementation will not explore this to avoid introducing bias to the results.

It is also important to note that unlike in some Neural-Linear [33, 19] bandit approaches – which this hybrid RS architecture is arguably in the vein of due to the use of *deep representations* and a much more shallow exploration layer invoking a linear bandit on the output embeddings – there will not be a focus here on a continually changing (non-stationary) embedding landscape, as this blurs this line of work through additional considerations. Stochastic linear bandits take environment stationarity as a central assumption [27], requiring particular adaptation that would

affect the generalisability of this model's results; such work is beyond the scope of this project. It is a crucial framing that this is the study of bandits within a fixed frozen embedding window, analysing the supplementary capabilities of the bandit in guiding the model during its period of feedback delay in which it cannot immediately retrain its entire embedding space on real-time user interactions.

The central accuracy metric will be that of *cumulative reward* based on the aforementioned binary signal, observed across all four algorithms for all time steps. This matches the standard evaluation metric for bandit algorithm accuracy in offline contexts, and has seen specific use in the evaluation of bandits on the KuaiRec dataset [32]. Following the direction of that paper, this reward for each time step will be averaged across all simulated user streams, to debias the results from the order of user interactions.

For the beyond-accuracy metrics, it will be crucial to not constrain the evaluation of exploration to a limited perspective that only looks at one measure of bias; work on diversity, serendipidity and fairness as separate metrics [11] will be used to inform the evaluative process. Diversity is a measure of item dissimilarity, for which the two-tower model's robust notion of similarity is incredibly useful; serendipidity is a measure of an item being unexpected and unfamiliar; and fairness is a broader look at the reduction of specific user and item biases [6]. The fully-exposed nature of the KuaiRec dataset, as well as access to high-quality user and item embeddings alongside a similarity function, provides great opportunity to devise an interpretable, high-quality evaluation of the bandits' exploration; this is left to later in the project if time permits, as focus must be placed on ensuring that the essential elements of this project are delivered.

Regardless, given the central requirement noted in the original specification (Appendix A) and throughout this project of a robust measure of *exploration quality*, there are non-negotiable key beyond-diversity metrics to guide exploration evaluation: the *Gini coefficient* (as a measure of inequality in item recommendation frequencies), long-tail exposure frequency (as a means of measuring neglected items directly), and Average Recommendation Popularity (ARP) [6].

3.3 Outline of Remaining Work

Remaining work centres around the thorough implementation of this system design. Open-source PyTorch implementations of two-tower will be used to speed-up development [39] as many implementations can be found online; similarly for other system components such as the contextual bandits. Additionally as mentioned in the system design, there are areas that – given the rate of progress, the current status of the report, and noted room for improved evaluation

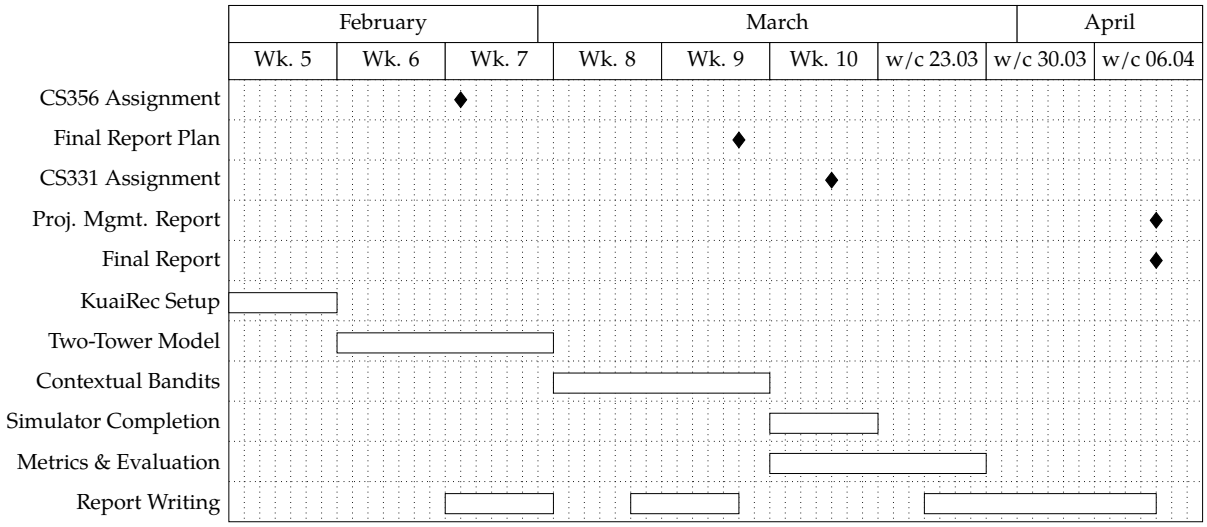


Figure 4: Proposed timetable for the rest of the project

and interpretability – additional design may be done to support discussion of the models' capabilities to balance the exploration-exploitation tradeoff. This may be in the inclusion of further beyond-accuracy metrics [6]; a more comprehensive look at bias [40]; or in visualisation that may be useful for interpretability in the report, such as lower dimensional visualisation of the shared embedding space. Nonetheless, work will focus primarily on meeting the system design first and ensuring a comprehensive final report and these are additional considerations.

Figure 4 outlines the planned timeline for fulfilling this – the time estimates are on the conservative side such as to cater for potential delays or issues in implementation. For example, the contextual bandits may be mathematically complex but on an implementation level they are not nearly as difficult as the two-tower embedding model, but it is expected their integration with KuaiRec's *small matrix* as part of the simulator will not be immediately simple. Additionally, report writing must be performed throughout to ensure that it is not all left to the very end of the project, and as to allow continual editing of the report – including supervisor feedback and the *plan* deadline on the 13th of March. Report writing also acts as a means of continually gauging progress and prompts wider understanding of the system design as it is implemented.

4 Reflections and Methodology Considerations

My general assessment of the project is that while I am incredibly relieved to have completed both the system design and to have researched thoroughly to discuss its architectural decisions, this process did take significantly longer than expected and that as a result a lot of pressure has been placed on the rest of the project's timescale. Due to significantly less teaching time and courseworks this term for my modules, however, I am optimistic.

A major flaw of my methodology so far has been that of not meeting with my supervisor enough during times of uncertainty – when stuck in grasping the literature, I often felt that I did not want to misrepresent the desired direction of the project in meetings nor my concerns, and continually felt that I needed to perform more research until I was better prepared to explain the entire system design to Prof. Florin Ciucu. This was incredibly naive in hindsight, especially in places where I did not completely grasp the deep-learning or bandit concepts, and for the rest of the project I will aim to continually involve my supervisor as part of my deliberation process. Given the busy nature of my supervisor's schedule, and difficulties in completely explaining the entire complexity of the project in-person, I will use emails to more formally outline long-standing concerns; this also should have been done in continually informing my supervisor of my personal circumstances.

Additionally, the writing of this report was incredibly helpful in cementing the system design as I had a large amount of research spanning many ideas but had yet to synthesise this work, outline the bridging processes (for example how two-tower embeddings become context vectors for the bandit), and draw clear lines on scope in a measured way to resolve ambiguities. I had gauged prior that I was not in the position to resolve these ambiguities and that they would be resolved iteratively when coding as I would gain understanding of these systems; I have learned that holistic up-front system planning is crucial for this research-heavy project due to the high cost of spending weeks researching and implementing subsystems, as opposed to a purely iterative methodology, which is only now suitable given much clearer software deliverables.

For the software, I am using Git both as a means of managing an iterative approach through development branches, and for rigorous version control in the case of any major issues with the system; this is continually pushed remotely to GitHub to ensure that the files are backed up.

5 Legal, Social, Ethical and Professional Considerations

No special social, ethical, nor professional considerations have been identified for this work: publicly available, purely anonymised user-item features will be used from KuaiRec. From a legal standpoint its CC-BY-SA 4.0 license will be respected in distributing my trained model under the same license if made public alongside thorough attribution.

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Appendix

A Project Specification

CS310 Project Specification

Bandits for Online Exploration in Deep Recommenders

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Abstract

An investigation into multi-armed bandits (MABs), specifically generalisations such as the contextual bandit and their solutions, for alleviating existing policy bias and encouraging exploration in modern deep-learning recommendation systems.

1 Problem outline & this project

1.1 Modern deep-learning recommendation systems

Recommendation systems are crucial for serving personalised experiences in the modern world across massive user and content data sets, with major uses spanning e-commerce, social media, and streaming media services. In recent years, neural and deep-learning models for recommenders have become dominant, largely due to their efficiency in identifying complex non-linear relationships between users, interests, and potential recommendation items across incredibly large search spaces [1].

Limitations, however, often arise in these deep-learning models' abilities to properly explore new items and changing relationships in real-time, as they can become reliant on prioritising pre-established relationships and high-engagement items. Research shows [2, 3] that they often converge to suboptimal recommendation policies, hampering further learning.

As long as a recommendation is *good enough*, a feedback loop can occur in which the recommendation system continually recommends items that already have high engagement [3]. Additionally, these models are often reliant on batch learning, arguably lacking the

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design to facilitate exploration in real-time (*online*) learning despite being deployed in real-time environments due to model update latency.

1.2 The multi-armed bandit problem

The multi-armed bandit (MAB) problem is a reinforcement learning problem in which an agent has a set of fixed choices, with the reward behaviours of the choices not completely known prior; the agent has finite rounds to iteratively pick between the choices, gradually gaining knowledge of the reward distributions while aiming to maximise their cumulative reward.

MAB solutions must navigate a balance of exploration and exploitation to optimise reward, and so they have been employed in the design of online, real-time recommenders. Specifically, they often involve an important generalisation: the *contextual bandit* in which the agent also has access to context information, and the system learns a relationship between choice rewards and context. This allows for user-specific recommendations, given that user characteristics are effectively represented in the context vector. [4]

Historically, these recommender designs have struggled with the scale of real-world systems, especially in comparison to the efficiency of batch-trained neural models such as the *two-tower architecture* in reducing search spaces for recommendation. However, recent work has focused on successfully developing scalable bandit systems for online recommender learning [5], albeit still with noted opportunity for integration with offline training to reduce exploration overhead.

1.3 The demand for a hybrid approach

Given the recent developments in scalable online bandit infrastructure for exploration in recommenders, and the existing benefits of offline deep-learning techniques for handling search space reduction, there is opportunity for new work that combines them for an efficient, scalable hybrid *offline and online* approach. Recently, Yi et al. [6] has outlined such an approach, but with limited application and evaluation in-house at YouTube, and notably without consideration of newer scalable neural bandit approaches.

This project will explore the existing literature on offline deep-learning and online bandit recommender architectures, with the aim of developing a comprehensive hybrid approach

that supplements existing deep-learning recommenders with recent large-scale bandit algorithms for exploration and the mitigation of suboptimal policy feedback loops.

The success of such an approach will hopefully outline a potential avenue for the improvement of recommendation quality in industry-scale recommenders, with particularly important applications in cases such as closed social media content ecosystems where items neglected by recommender algorithms often lack alternative mechanisms to gain popularity, and so there are less external pressures on these recommenders to counter policy stagnation and insufficient exploration.

2 Objectives

The general objective of this project is the production of the aforementioned hybrid recommendation system design, supplementing an *offline* deep-learning recommender architecture with an *online* bandit algorithm for increased exploration, while maintaining performance and recommendation quality at scale. This can be concretely broken down as follows:

2.1 Essential

These are minimum requirements for achieving the goals of this project.

- Research *offline* deep-learning architectures used in large-scale recommendation systems, identifying candidate deep-learning components for the foundation of a hybrid recommender system design.
- Research *online* bandit architectures suitable for large-scale recommendation systems, identifying candidate bandit models for the exploration layer(s) of a hybrid recommender system design.
- Design and implement a hybrid recommendation system combining the selected offline deep-learning and online bandit recommender designs.
- Perform *scalability* (benchmark) testing of this hybrid recommender against existing deep-learning recommenders on large-scale datasets to analyse and demonstrate the comparable performance and resource viability of this approach.
- Perform *recommendation quality* testing of this hybrid recommender against existing

deep-learning recommenders on standard recommendation quality benchmarks (e.g. the MovieLens dataset) in both pre-trained and purely online contexts to ensure that the exploration system has not compromised overall recommender quality.

- Perform *real-time exploration* (online learning) testing of this hybrid recommender against existing deep-learning recommenders to assess the real-time learning improvements of this approach with respect to new or changing user-item relationships.
- Tweak the design and implementation of the hybrid recommender with respect to the results of the tests to ensure an appropriate tradeoff between real-time learning improvements, overall recommendation quality, and model scalability.

2.2 Extensions

Given successful achievement of the essential criteria, the following objectives outline additional avenues for potential work, but will not be the primary focus of work unless time permits, and are not necessary for project success.

- Research and implement a potential approach to spread the load of exploration across users (bandit contexts), as although increased exploration can improve long-term convergence towards optimal recommendation policies, it may cause immediate losses in user experience.
- Research and implement a potential approach to vary overall exploration in cases of observed user disinterest.

2.3 Beyond the scope of this project

The following outlines limits for the research scope of this project, to direct work and ensure maintained focus on the essential objectives within our timeframe.

- This project will not aim for scalability improvements in comparison to state-of-the-art recommender architectures, merely demonstrated comparable viability at scale.
- This project will not aim to innovate on and optimise existing deep-learning components of the recommendation pipeline, merely to adapt suitable architectures for

a hybrid recommender.

- This project will not aim to present a completely novel general-purpose bandit strategy, merely to adapt existing strategies for a hybrid recommender.
- This project will not consider recommender systems outside of offline deep-learning and online bandit architectures.

These limits reinforce the key focus of this project: demonstration of a viable hybrid offline-online deep-learning and bandit recommender system, with measured improvements in the ability of the recommender to explore new or changing relationships against existing deep-learning recommenders. The work of integrating these two systems effectively is a challenge enough, and tackling these limits could constitute a whole separate research project each.

3 Methods & Methodology

3.1 Project Management

As research, development, and testing must be continually undertaken to determine the feasible deep-learning and bandit components for this hybrid recommender system, as well as how to effectively bridge them, an iterative methodology is best suited for managing this project - a purely sequential approach risks an unfeasible or poor recommender design. There are, however, some plan-driven elements required due to the fixed report deliverables. As such, a hybrid approach that leans heavily into Agile principles, but with robust timeframes planned for documentation throughout the project, seems most appropriate for this project.

Work will be broken into two-week sprints, with a general aim of earlier sprints focusing on researching and experimenting with constituent deep-learning and bandit subsystems suitable for the hybrid design, and later sprints gradually designing and bringing together those subsystems. Research, development, and testing will be interwoven as mentioned prior, to guide feasibility of the final design. Partial informal documentation will occur alongside all sprints, but specific sprints will be planned ahead of time to involve formal work on the documentation due to the fixed report deliverables. Project supervisor meetings will occur at least bi-weekly, but may occur weekly where new ideas arise.

3.2 Research, development & evaluation

This project will require the aforementioned research of deep-learning recommender architectures, bandit recommender architectures, and potential approaches to bridging these systems. This will be done primarily through extensive literature review, but also through the study of any open-source repositories, tutorials, and textbooks as noted further in Resources.

Development will involve the technologies as outlined in Resources. Git will be used for version control, with branches used to aid the iterative development methodology in testing and continually integrating candidate recommender components. The repository will be hosted on GitHub, with frequent remote pushes to ensure that progress is not lost in the case of hardware failure.

Evaluation will involve a test suite developed that fulfils the comprehensive testing requirements of the Essential Objectives: (1) demonstration that the recommender retains comparative scalability and recommendation quality, (2) demonstration that the recommender has a comparative improvement in exploration ability, both with respect to existing deep-learning recommenders. Specific datasets will depend on the nature of the final hybrid system, but candidates are outlined in Resources, and can be split up for online simulation. Attention will be paid to randomising the training data arrival order across different tests, with seeds for reproducibility. Testing against existing models will involve open-source implementations where possible, of which there are many available online for various different recommender architectures.

4 Timetable

Period	Week	Sprint focus
Term 1	1 2	**Project Specification**
	3 4	Literature review
	5 6	Iterative prototyping of deep-learning & bandit subcomponents

	7 8	Iterative prototyping of combined hybrid system
	9 10	Initial development of test suite
Christmas Holiday	All	**Progress Report** & flexible buffer period
Term 2	1 2	Refinement of hybrid system design
	3 4	Refinement of testing suite
	5 6	Refinement of hybrid system in response to tests
	7 8	Completion of final testing suite
	9 10	**Final Report** & **Project Management Report**
Easter Holiday	All	**Final Report** & **Project Management Report**

Table 1: A general timeline for the project

As discussed prior, this project will employ a hybrid methodology due to the plan-driven pressures of fixed report deadlines, and as such *general outlines* for sprint focuses are given above. These two-week sprints, however, will operate using an iterative process of interwoven research, development, and testing. Testing will involve small toy examples to ensure basic feasibility and robustness at early stages until development of the test suite.

5 Resources & Risks

5.1 Resources

Research resources will include but are not limited to:

- existing deep-learning and bandit recommender literature
- the *Online Matching* [6] hybrid system outline
- online resources for implementing recommender systems

- available open-source recommender implementations
- relevant textbooks (including *Bandit Algorithms* [7])

Development will be in `Python` using the `PyTorch` library for deep learning, given its ease of use and high-quality documentation, and the `NumPy` library for large-scale matrix calculations required for both the deep-learning and the bandit components. For hardware, my laptop and my desktop (GPU: 2080TI, 11GB VRAM) will suffice for developing and running the recommender on small toy datasets; for larger-scale testing and model training, I will use the DCS Batch Compute System.

Testing will involve standard datasets used for evaluating recommender systems e.g. MovieLens (movies), Amazon (e-commerce), LastFM (music), MIND (news), MicroLens (short-form video); final dataset choices will depend on those used in literature for testing the chosen hybrid system subcomponents.

5.2 Risks

A primary risk is that of hardware failure; mitigation to ensure that the recommender codebase is not lost has been outlined in the development methodology (i.e. continual pushes to GitHub). If I am unable to continue work on my own machines, I will be able to use the DCS machines without issue, and if DCS machines and/or the Batch Compute System are unavailable I can use Google Colab temporarily for free.

Another major risk is that of the project not being feasible as currently formulated, research taking too long, development and/or testing proving difficult, and other project management risks. This has been accounted for through the use of an Agile-driven methodology to ensure a robust, iterative approach. Progress and remaining goals will be assessed at the end of each sprint, and project scope will be reduced or altered as required (and possible) if the project is deemed unfeasible or is found to ultimately have existing solutions of exactly the same approach or better. Buffer time is provided in the holidays to allow the project extra time to bounce back from potential issues.

6 Legal, Social, Ethical & Professional Issues

No social, ethical or professional issues were identified. The datasets used for evaluation are both publicly available and anonymised, and so do not require ethical consent. From a legal standpoint, usage license terms will be respected for all datasets.

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